Weather Monitoring System

Aim : To design and develop a weather monitoring system using ESP32-S3 microcontroller that collects environmental data from multiple sensors, transfers the data through wireless communication, processes it in software, stores the readings, and displays real-time information through a dashboard.

Objectives :

- To collect environmental data using ESP32-S3 with BME280 and PT100 sensors.
- To transmit sensor readings through WiFi communication.
- To process, store, and analyze sensor data using software modules.
- To display real-time data and alerts through a PyQt6 dashboard.

Sensors :

BME280 :

Parameters measured:

- Pressure
- Humidity

Communication:

- SPI Interface
- The ESP32-S3 acts as the master device and collects sensor values from BME280.

PT100 :

Parameter measured:

- Temperature

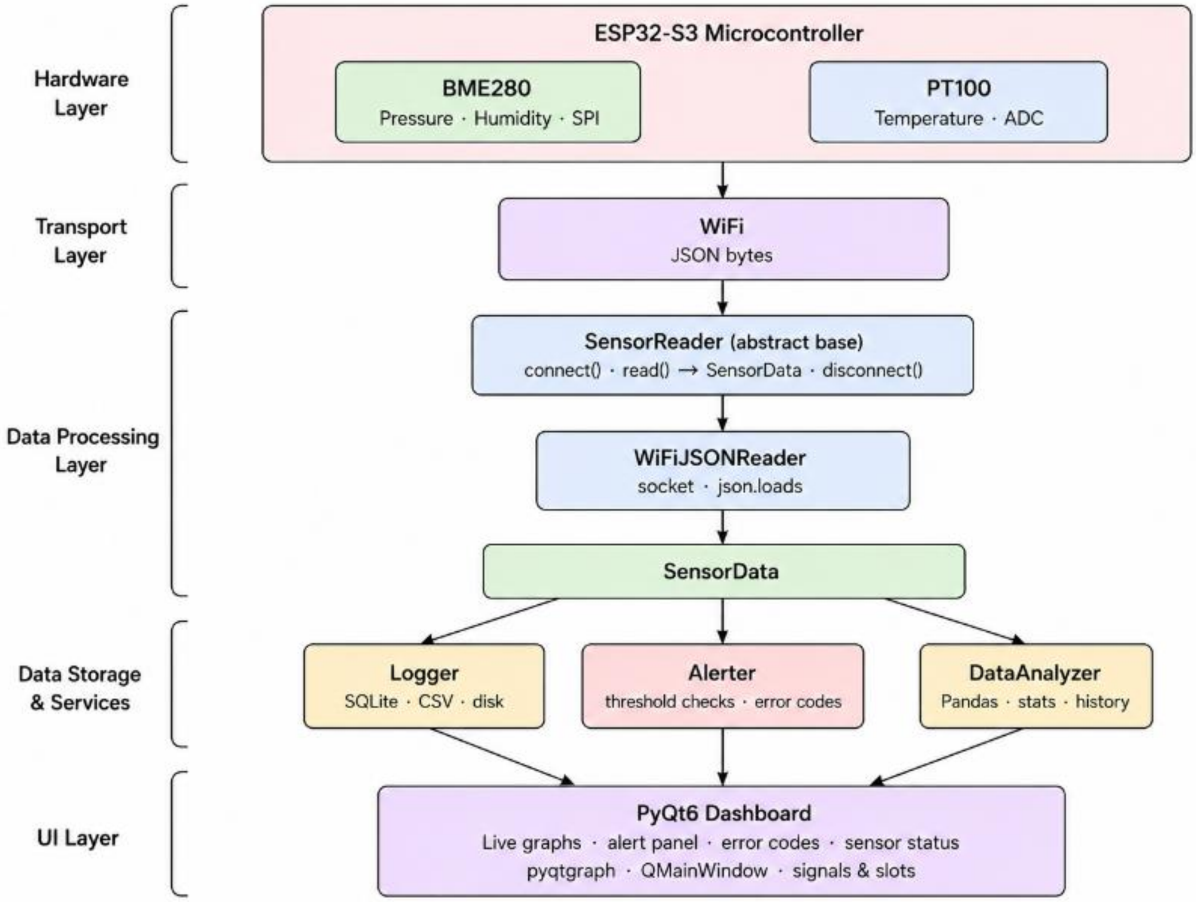
PT100 is an RTD sensor whose resistance changes with temperature.

Approach :

The weather monitoring system uses an ESP32-S3 microcontroller to collect sensor data from BME280 and PT100 sensors. The collected readings are transferred to the software application through Wi-Fi communication.

The SensorReader module receives the data, where WiFiJSONReader parses the incoming JSON packets and converts them into a SensorData object. The parsed data is stored using Logger, monitored through Alerter, and analyzed using Data Analyzer.

Finally, the PyQt6 dashboard displays real-time sensor values, graphs, sensor status, and alerts.



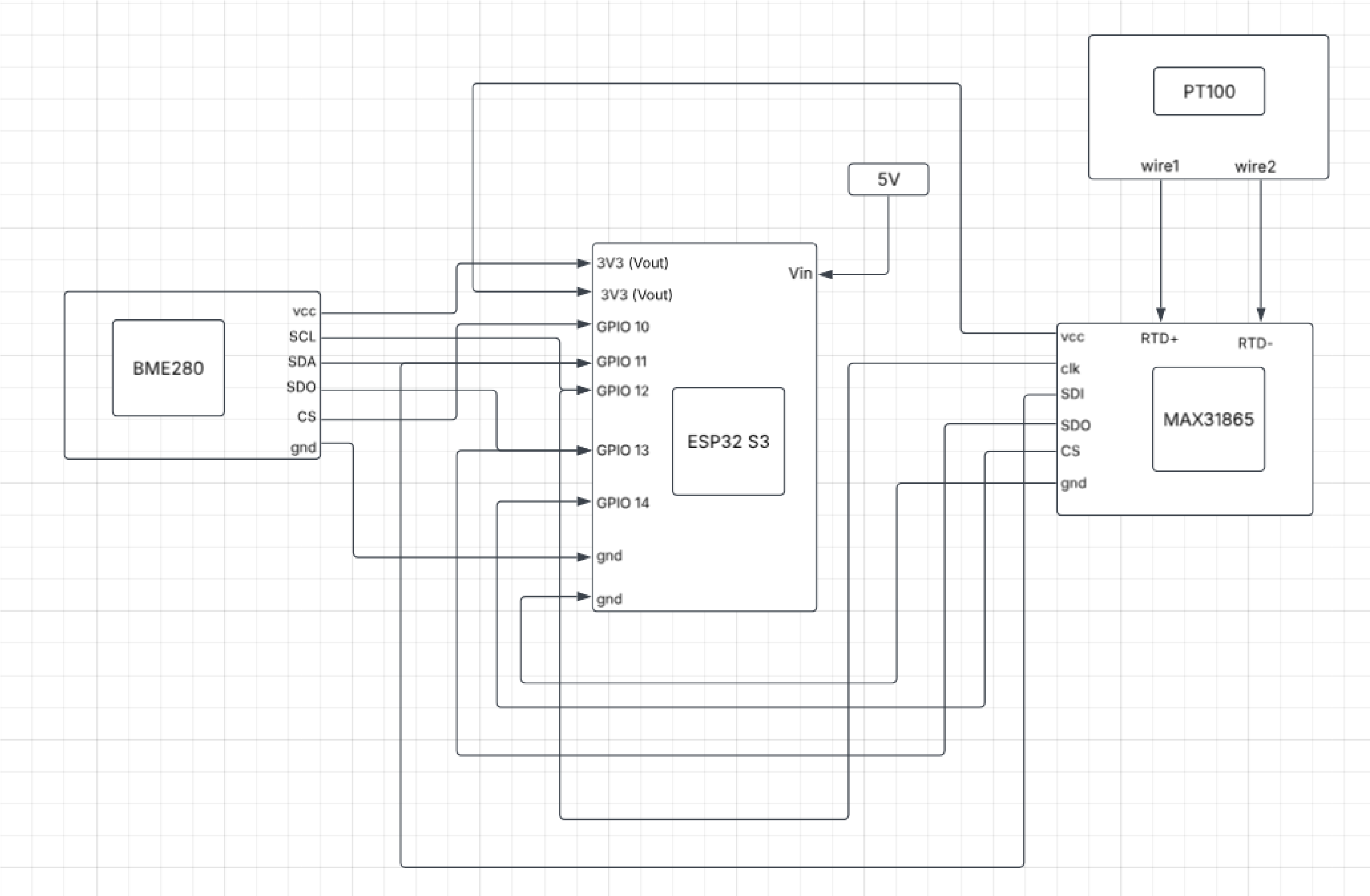
Hardware connections :

The ESP32-S3 connects to both sensors using the SPI communication protocol, which allows fast data transfer and supports multiple devices on a single bus through separate chip-select (CS) lines.

The BME280 and MAX31865 share the same SPI bus (CLK, MOSI, MISO), but each has its own dedicated CS pin. This lets the ESP32-S3 select which sensor to communicate with at a time, avoiding data conflicts.

The BME280 directly sends pressure and humidity data over SPI. The PT100, however, only produces a raw resistance value, so it cannot connect directly to the ESP32-S3. The MAX31865 handles this by reading the PT100's resistance and converting it into a temperature reading, which is then sent to the ESP32-S3 over the shared SPI bus using its own CS line.

Both sensors are powered through the ESP32-S3's 3.3V output, while the microcontroller itself runs on an external 5V supply.



Tools and Technologies Used :

Hardware:

- ESP32-S3 Microcontroller
- BME280
- PT100
- MAX31865

Software:

- Python
- Arduino IDE / ESP-IDF
- PyQt6